

Context Engineering 4-Layer Architecture

Data source: [Liu et al., 2023] [Zheng et al., 2024]

Layer 1: Content Organization

Key info at start and end · Minor info in middle · Lost in the Middle

Position Optimization



Layer 2: Context Compression

Summary compression · Selective retention · Prompt caching · RadixAttention

Length Optimization



Layer 3: Dynamic Loading

On-demand retrieval · Relevance evaluation · Timeliness judgment · Memory integration

Content Optimization



Layer 4: Token Budget & Monitoring

Cost ceiling · Output length limit · Anomaly detection

Cost Control